ADAPTIVE STAIRCASE
PSY310: Lab in Psychology
Lab Report

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GITHUB LINK:

Introduction

In this experiment, the focus is on the connection between a physical stimulus and a human being's sensation through the study of psychophysical thresholds and their relationship with the rest of the sensory modalities. A threshold is an interval within which a stimulus is undefinably weak and yet capable of being sensed. These are termed absolute and difference thresholds – absolute being the minimum stimulus intensity which can be sensed, while a difference threshold (or a Just Noticeable Difference) is the lowest change in stimulus intensity that can be sensed. Traditional approaches to threshold assessment, such as the method of limits or the method of constant stimuli, are tedious, and not the most resourceful. The adaptive staircase method is a more suitable alternative. This is more popularly referred to as "the method of ups and downs." The adaptive staircase method is more efficient because it alters the stimulus intensity proportionate to the participants responses, resulting in a more accurate threshold. This is a common procedure in vision and audiology research. The objective of this study was to The difference threshold for the discrimination of orientation was measured using the adaptive staircase procedure.

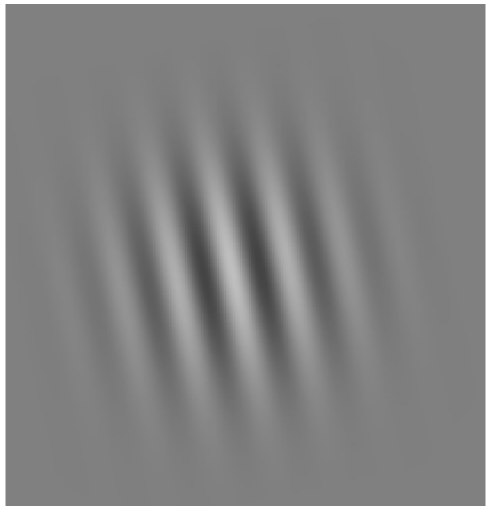
Method

Participants: The participant was a 20-year-old undergraduate student from Ahmedabad University.

Materials and Stimuli: The experiment was created using PsychoPy v2025.1.1, a popular Python-based software for psychophysics research. It was run on a 13-inch monitor with a resolution of 1920 x 1080 pixels. The visual stimulus was a Gaussian Mask with a sinusoidal texture. Its settings included a spatial frequency of 8, a texture resolution of 128, and linear interpolation, with a contrast of 0.5. A cross-shaped figure was used for fixation and was displayed for 0.5 seconds before each trial.

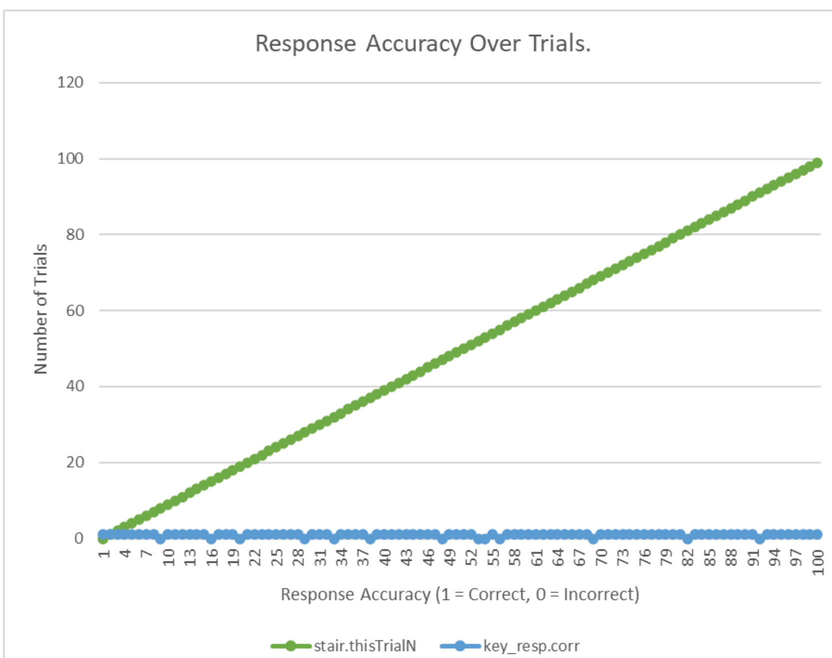
Procedure: The experiment consisted of 100 adaptive staircase trials. The initial orientation of the lines was set to 10 degrees, with lines randomly tilted to the left or right. The maximum possible orientation was 30 degrees, and the minimum was 1 degree. A 1-up, 3-down adaptive staircase procedure was used, with step sizes of [2, 1, 1, 0.5] and a linear step type.
The participant's task was to press the left arrow key (⟵) if the grating lines were tilted to the left, and the right arrow key (⟶) if they were tilted to the right. PsychoPy's built-in features automatically recorded data for each trial, including the line tilt and the participant's response. At the end of the experiment, an Excel file was generated, which was then used to analyze the difference threshold.

(Image 1: A Gaussian Mask tilted to the left side)



Results

To understand the participant's perception of the minimum stimulus intensity they could detect, the average of the last five intensity thresholds was calculated. The last five thresholds from the 100 trials, where the average intensity value was observed to be 1.2 deg. Hence, the absolute threshold was 1.2 deg (Figure 1).



(fig. 1: A line chart showing the participant's responses over 100 trials, where the absolute threshold was determined to be 1.2 deg.)

Additionally, the participant demonstrated an overall accuracy rate of 72% throughout the 100 trials. This was derived from the total number of correct responses divided by the total number of trials. The last five reversals were chosen to calculate the absolute threshold, as they provide a more stable and realistic representation of the participant's performance after they have become accustomed to the experimental procedure.

Discussion

The analysis of the participant's responses implied a threshold value of 1.2 deg with a 72% accuracy. The lowest intensity at which the participant could consistently detect a change in stimuli is 1.2 degrees. Given the low threshold, this suggests that the individual is reasonably responsive to the stimuli. However, the 72% accuracy rate suggests that the participant's sensitivity was high but not flawless. It should be noted that there may have been some fluctuation in the responses due to other factors (fatigue, inattention, etc.). These biases may compromise the experiment's internal validity, even though the adaptive staircase method is a useful strategy for rapidly identifying a threshold. The presence of these variables suggests that more research will be required to ensure the external validity and reliability of the threshold value, perhaps by rerunning the experiment with other conditions or with a bigger sample size.

References

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